

1)A 4-bit serial-in parallel-out shift register is initially set to 1111. The data 1010 is applied to the input. After 3 clock cycles the output will be:

- a)1100
- b)0101
- c)1011
- d)1010

The correct answer is

0101

solution

Step-by-Step Shift Register Operation

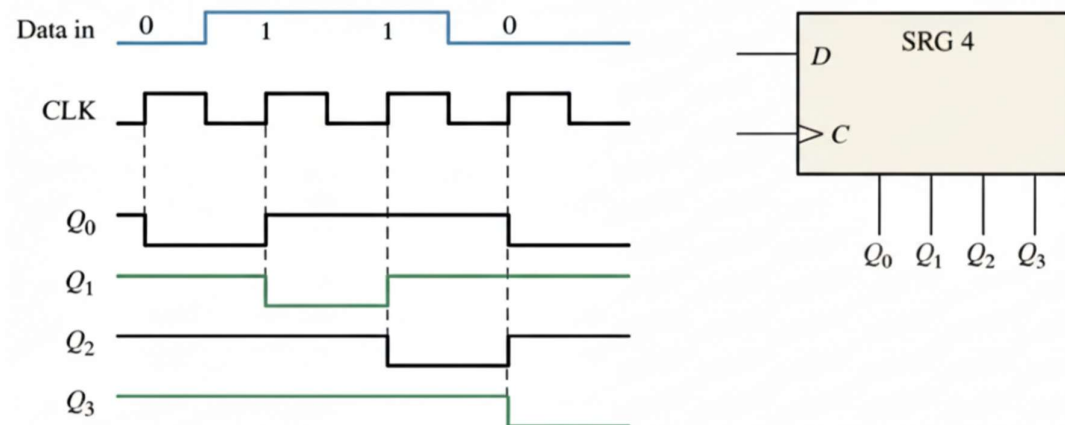
With each clock pulse, a new serial input bit enters Q_3 , and the bit from Q_3 moves to Q_2 , Q_2 to Q_1 , and Q_1 to Q_0 . The bit that was in Q_0 is shifted out of the register and effectively lost.

Clock Cycle (T)	Serial Input (D_{in})	Q_3 (MSB)	Q_2	Q_1	Q_0 (LSB)	Register State ($Q_3Q_2Q_1Q_0$)
Initial (T=0)	-	1	1	1	1	1111
1	0 (LSB of 1010)	0	1	1	1	0111
2	1 (next bit of 1010)	1	0	1	1	1011
3	0 (next bit of 1010)	0	1	0	1	0101

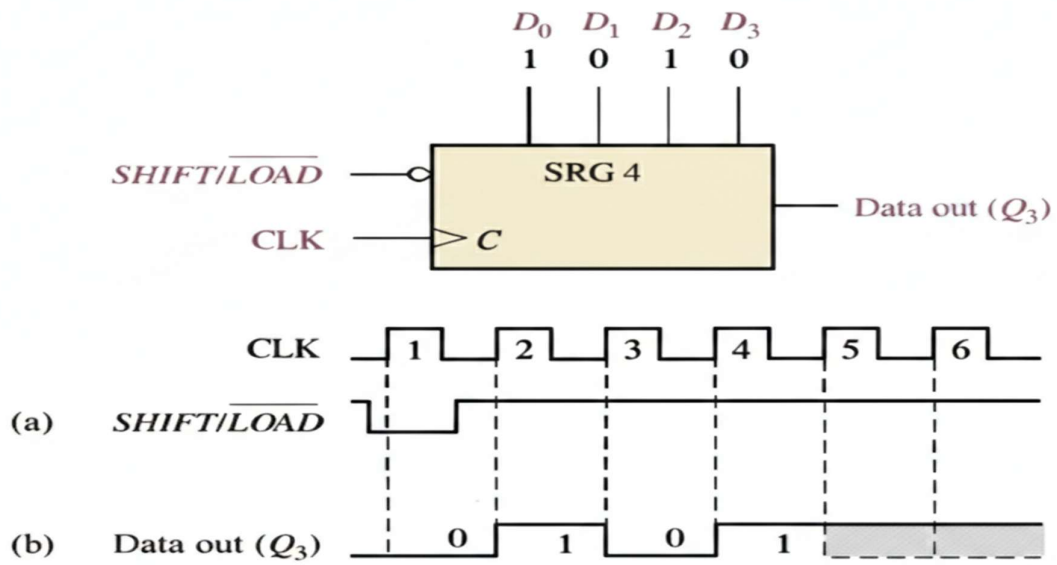
2) Show the states of the 4-bit SIPO register (SRG 4) for the data input 0110 and clock waveforms. The register initially contains all 1's.

Solution

The register contains 0110 after four clock pulses. .



3) Draw the data-output waveform(Q_3) for a 4-bit PISO register SRG4 shown below for 1010 input.



On clock pulse 1, the parallel data ($D_0D_1D_2D_3 = 1010$) are loaded into the register, making Q_3 '0'. On clock pulse 2 the '1' from Q_2 , is shifted onto Q_3 ; on clock pulse 3 the 0 is shifted onto Q_3 ; on clock pulse 4 the last data bit (1) is shifted onto Q_3 ; and on clock pulse 5, all data bits have been shifted out, and only 1's remain in the register (assuming the D_0 input remains a 1).